

Final Project – SQL Code Screenshots

03_analysis.sql

Figure 03-1

Analysis 1: Total Sales per State

shipping_state	total_orders	total_revenue
FL	25	38327.04
PA	13	15921.61
CA	11	14161.88
NY	9	11016.00
IL	6	6715.46
MA	6	6272.18
AZ	3	4712.73
WA	3	3879.62
TX	2	2329.51
GA	2	1506.90

Figure 03-2

Analysis 2: CTE and Window Function - Rank States by Per-Capita Sales

shipping_state	region	state_revenue	population	sales_per_capita	per_capita_rank
FL	South	38327.04	21538187	0.0018	1
PA	Northeast	15921.61	13002700	0.0012	2
MA	Northeast	6272.18	7029917	0.0009	3
AZ	West	4712.73	7276316	0.0006	4
NY	Northeast	11016.00	19835913	0.0006	5
IL	Midwest	6715.46	12812508	0.0005	6
WA	West	3879.62	7705281	0.0005	7
CA	West	14161.88	39538223	0.0004	8
GA	South	1506.90	10711908	0.0001	9
TX	South	2329.51	29145505	0.0001	10

Figure 03-3

Analysis 3: Advanced Aggregation - Revenue vs. Population Segments

population_segment	state_count	avg_revenue_per_state	avg_urban_pct
High Pop (>10M)	7	12854.06	86.0
Med Pop (5M-10M)	3	4954.84	88.6

Figure 03-4

Analysis 4: Rolling Average Analysis - Using Window Function AVG

sales_month	monthly_sales	3_month_moving_avg
2022-01	2900.28	2900.280000
2022-02	2987.26	2943.770000
2022-03	6610.56	4166.033333
2022-04	2225.42	3941.080000
2022-05	3741.77	4192.583333
2022-06	4414.28	3460.490000
2022-07	2449.99	3535.346667
2022-08	2582.43	3148.900000
2022-11	3130.20	2720.873333
2022-12	3552.28	3088.303333
2023-01	4072.20	3584.893333
2023-02	5520.08	4381.520000
2023-03	7732.50	5774.926667
2023-04	1316.61	4856.396667
2023-05	2232.19	3760.433333
2023-06	4155.75	2568.183333
2023-07	1236.53	2541.490000
2023-08	4178.79	3190.356667
2023-09	2724.18	2713.166667
2023-10	5667.38	4190.116667
2023-11	4601.44	4331.000000
2023-12	2813.65	4360.823333
2024-01	3191.55	3535.546667
2024-03	761.76	2255.653333
2024-04	1583.64	1845.650000
2024-05	1720.19	1355.196667
2024-06	3186.52	2163.450000
2024-08	6229.20	3711.970000
2024-10	3909.70	4441.806667
2024-11	2397.16	4178.686667
2024-12	1017.44	2441.433333

Figure 03-5

Analysis 5: Regional Performance Comparison

region	total_orders	total_revenue	avg_order_value
South	29	42163.45	1453.91
Northeast	28	33209.79	1186.06
West	17	22754.23	1338.48
Midwest	6	6715.46	1119.24

Figure 03-6

Analysis 6: Impact of Income on Spending

income_tier	num_orders	avg_order_value
Middle Income (60k-75k)	71	1333.68
High Income (>75k)	9	1127.98

04_viz_prep.sql

Figure 04-1

A master view integrating all analytical dimensions

region	state	population	median_income	urban_pct	total_sales	num_orders	sales_per_capita	avg_order_value
South	FL	21538187	62000	91.2	38327.04	25	0.00178	1533.08
Northeast	PA	13002700	63000	78.7	15921.61	13	0.00122	1224.74
Northeast	MA	7029917	85000	92.0	6272.18	6	0.00089	1045.36
West	AZ	7276316	62000	89.8	4712.73	3	0.00065	1570.91
Northeast	NY	19835913	72000	87.9	11016.00	9	0.00056	1224.00
Midwest	IL	12812508	72000	88.5	6715.46	6	0.00052	1119.24
West	WA	7705281	77000	84.0	3879.62	3	0.00050	1293.21
West	CA	39538223	75000	94.2	14161.88	11	0.00036	1287.44
South	GA	10711908	60000	77.0	1506.90	2	0.00014	753.45
South	TX	29145505	64000	84.7	2329.51	2	0.00008	1164.76

Figure 04-2

Output for comparative analysis

region	region_total_sales	region_population	region_per_capita
South	42163.45	126296072	0.00033
Northeast	33209.79	57243812	0.00058
West	22754.23	78713578	0.00029
Midwest	6715.46	68985454	0.00010